Composites Science and Technology 216 (2021) 109068

Contents lists available at ScienceDirect

Composites Science and Technology

journal homepage: www.elsevier.com/locate/compscitech

Highly flexible and recyclable F–SiO_2/MPU composites for self-powered active motion sensors

Xu Xu [1], Yuting Ouyang [1], Shaorong Lu, Hong Ruan, Yuqi Li [*]

Key Laboratory of New Processing Technology for Nonferrous Metals and Materials, Ministry of Education, College of Materials Science and Engineering, Guilin University of Technology, Guilin, 541004, China

ARTICLE INFO

Keywords:
Recyclable
Polyurethane
Diels-alder
Triboelectric nanogenerator

ABSTRACT

Triboelectric nanogenerator (TENG) with flexibility, recyclability, and high mechanical performance are highly desirable for self-powered active motion sensors. Herein, maleic anhydride and furan groups were introduced into the polyurethane and SiO_2, respectively, to obtain furan modified SiO_2 (F–SiO_2) and maleic anhydride modified polyurethane (MPU). The dynamic crosslinked F–SiO_2/MPU composites (FM) possessed the recyclability due to the Diels-Alder reaction. In addition, the mechanical properties of FM were improved due to the improvement of the interface compatibility between F–SiO_2 and MPU. Especially, at 5 wt% content (FM-5), the tensile strength and elongation at break are 7.09 MPa and 845.61%, respectively. Compared with pure MPU, it has increased by 249.26% and 176.30%, respectively. Additionally, at 3 wt% F–SiO_2 content, the short-circuit current (*Isc*), open circuit voltage (*Voc*), transferred charge (*Qsc*) and peak power density of FM-3 increased by 18.60%, 36.04%, 49.24% and 13.00%, respectively. This work offers new insights into the design and preparation of triboelectric nanogenerator for self-powered active motion sensors with flexibility, and recyclability.

1. Introduction

In recent years, electronic equipment has shown the development trend of miniaturization, multi-function and low power consumption [1–3]. Personal portable electronic products such as laptop computers and mobile phones are rapidly popularized, which make it particularly important to solve small-scale electricity problems [4]. The current technical solution is mainly batteries. Unfortunately, the batteries themselves have many shortcomings, such as lack of flexibility, poor durability of power supply, serious environmental pollution and high consumption of material resources [5]. Therefore, a small and environmentally friendly new power generation equipment with high energy conversion rate is in urgent need of invented [6].

The triboelectric nanogenerator (TENG) is an energy harvesting device that collects different kinds of mechanical energy from the environment and converts it into electrical energy based on the combined action of the principles of triboelectricity and electrostatic induction [7–13]. TENG has the advantages of environmental friendliness, high energy conversion rate, small driving force required and diversified energy sources [14,15]. Therefore, one of the most promising application area of TENG is self-powered sensors. Due to the huge number of sensor devices and the wide range of applications, it is impractical to use traditional power supply methods (such as rigid batteries). TENG can be used as a self-powered sensor to monitor movement, vibration, human triggers, and object contact [16,17]. Compared with traditional sensors that use batteries for power supply, TENG has many advantages. Firstly, there are many kinds of materials suitable for making TENG, and we can choose materials with relatively low discomfort to the wearers. Secondly, TENG can convert the mechanical energy of motion into electrical energy, which can conveniently monitor human behavior in real time through electrical signals. The most important thing is that TENG is a self-powered sensor, so no external power supply is required [18]. Taking the above factors into consideration, TENG has broad development prospects in the sensor field.

At this stage, most materials for TENGs reported cannot be both flexible and stretchable while also being recyclable. What's more, as environmental pollution and waste of resources become more and more serious around the world, the recyclability of materials has become a hot topic for scientists. As is known to all, thermoplastic materials have

* Corresponding author.
E-mail address: liyuqi@glut.edu.cn (Y. Li).
[1] These authors contributed equally to this work.

https://doi.org/10.1016/j.compscitech.2021.109068
Received 3 May 2021; Received in revised form 15 August 2021; Accepted 24 September 2021
Available online 29 September 2021
0266-3538/© 2021 Elsevier Ltd. All rights reserved.

worse mechanical properties and thermal properties than thermosetting materials. Thermosetting materials cannot be reused. Therefore, the key to preparing a high-performance polymer is to have the high-strength and heat-resistant properties of thermosetting materials and the repeatability of thermoplastic polymers. Under these circumstances, the dynamic transition between thermosetting polymers and thermoplastic polymers is a research hotspot. Diels-Alder (D-A) reaction is a dynamic covalent chemical reaction that is temperature reversible [19,20]. At low temperature, the electron-rich conjugated diene structure reacts with the electron-deficient monoolefin to form a cyclohexene structure; at high temperature, the cyclohexene structure will be destroyed, regenerating conjugated diene and monoene. D-A reaction is expected to be introduced into polymer systems to realize the dynamic transformation of thermosetting and thermoplastic polymers [21]. Watanabe et al. used polyhexylene adipate to prepare furfuryl alcohol-terminated compounds. They used difunctional and trifunctional maleimide cross-linkers to prepare linear and bulk polymer materials, respectively. The material can undergo multiple cycles without changing the molecular weight and mechanical properties [22]. Gousse et al. used furfurylamine and maleic anhydride to react to synthesize small monomers with furan units and maleic anhydride units at both ends, which was used to synthesize linear polymer materials [23]. In summary, we can use the reversibility of the D-A reaction to prepare a new type of recyclable TENG material, which can promote the development of environmental protection.

Herein, in order to obtain a thermosetting and thermoplastic reversible transformation materials for TENG, the maleic anhydride and furan ring were introduced into the polyurethane chain and the surface of SiO_2, respectively. Utilizing the D-A reaction, the F–SiO_2/MPU(FM) composites possess recycle processing properties. The original FM films were dissolved in DMF and vacuum-dried again to obtain recycled films. By testing the mechanical properties and electrical properties of the recycled films and the original films and comparing the values, it can be fully demonstrated that the FM films have excellent recyclability. In addition, the interface compatibility between MPU and F–SiO_2 was improved, as well as the output properties of FM-TENG were increased. Hence, the FM composites have great potential as flexible, recyclable and self-powered sensors.

2. Material and methods

2.1. Materials

Silicon oxide (SiO_2) was obtained from Beijing Innochem Sci and Tech. Co., Ltd., (Beijing, China). Furfuryl alcohol and toluene were purchased from Aladdin Reagent Co. (Shanghai, China). Furan (GC >99.0%), ditin butyl dilaurate (DBTDL), ethyl acetate, absolute ethanol (EtOH) and maleic anhydride were acquired from Beijing Innochem Sci and Tech. Co., Ltd., (Beijing, China). Polycaprolactone diol (PCL, average Mñ1000) was acquired from Sigma-Aldrich Trading Co., Ltd., (Shanghai, China). 3-Amino-1,2-propanediol (APO, purity: 98%) and Hexamethylene diisocyanate (HDI, purity: 99%), N, *N*-Dimethylformamide (DMF) were also purchased from Beijing Innochem Sci and Tech. Co., Ltd., (Beijing, China).

2.2. Furan-SiO_2 (F–SiO_2) preparation

0.5 g SiO_2 and 50 ml DMF were added into a three-necked flask for ultrasonic dispersion for 1 h. Then, 5 g HDI and 0.25 ml DBTDL (catalyst) were added into the three-necked flask. After the reaction has been carried out at 80 °C for 1 h under the protection of nitrogen (N_2), 3 g furfuryl alcohol was added. The whole system continued to react at 80 °C for 1 h under the protection of N_2. Finally, the product in the flask was suction filtered and vacuum dried to obtain furan modified silicon oxide (F–SiO_2).

2.3. Synthesis of polyurethane

Firstly, the chain extender *N*-(2,3-dihydroxypropyl) maleimide (DHPM) was synthesized according to Yu's essay [24]. Then, the polyurethane was synthesized as follow: First, 5 g PCL that had been vacuum dried at 100 °C for 12 h was added to the flask, and the temperature was set to 75 °C. Second, add 0.025 ml DBTDL (catalyst) and 1.682 g HDI to the flask. The reaction has been carried out at 75 °C for 2 h. Finally, 0.855 g DHPM was added to the system to continue the reaction for 4 h. After the reaction, 50 ml DMF was poured into the flask and stirred at 130 °C for 0.5 h. The obtained solution was poured into a tetrafluoromethane mold and dried under vacuum at 100 °C for 24 h to obtain MPU.

2.4. F–SiO_2/MPU (FM) composite preparation

The preparation of FM-1 (F–SiO_2 content is 1 wt%) is discussed here to illustrate the procedure. First, 13 ml DMF and 1.485 g MPU were added to the flask. Second, the flask was heated to 130 °C. After the MPU is dissolved, 0.015 g of F–SiO_2 (1 wt%) was added. After stirring uniformly, ultrasonic dispersion was performed for 10 min. Then, the solution was quickly poured into a tetrafluoromethane mold and vacuum dried at 100 °C for 24 h. FM-1 was obtained after simple release treatment. FM-3 (F–SiO_2 content is 3 wt%), FM-5(F–SiO_2 content is 5 wt%), and FM-7(F–SiO_2 content is 7 wt%) composites were prepared according to the above experimental procedures.

2.5. Characterization and measurement

A fourier transform infrared spectra (FTIR) of F–SiO_2 and DHPM in wavenumber range from 500 to 4000 cm^{-1} were characterized by infrared spectrometer (Thermo Nicolet Nexus 470, USA) with using a KBr wafer. A TGA Q-500 thermogravimetric analyzer (TGA Q-500, TA Instrument, Brussels, Belgium) was used to test the thermal decomposition temperatures of the samples at a heating rate of 20 °C/min up to 700 °C. The tensile strength was evaluated on a universal tensile tester (UTM4503BLXY, Shenzhen Suns Technology Co., Ltd., Shenzhen, China) according to National Standard of China (GB1040-92) at the rate of 20 mm/min. The glass transition temperatures (T_g) of the samples were determined using a differential scanning calorimetry spectrometer (DSC-204, NETZSCH. Co., Ltd., Germany) at a heating rate of 20 °C/min under flowing N_2. An AVANCE-500 nuclear magnetic resonance spectrometer (AVANCE-500, Bruker company, Germany) was used to test the ^{1}H-NMR spectrum of the samples. The 6514 electrometer (6514, Keithley, USA) was used to test the open circuit voltage, short circuit current, and short circuit transfer charge amount of F–SiO_2/MPU-TENG. The working frequency during the test was 1 Hz.

2.6. Crosslinking density test

The degree of cross-linking of the cross-linked network structure of FM was characterized by testing the degree of swelling (Q) and gel content (G). The swelling degree (Q) was tested by swelling in chloroform at 25 °C for 24 h and weighing the mass of the samples before and after swelling. The gel content (G) test was to extract with chloroform in a Soxhlet extractor for 24 h to compare the quality of dry gel before and after extraction. The specific calculation formulas are shown in 2-1 and 2-2.

$$Q = 1 + \frac{\rho_1}{\rho_2}\left(\frac{m_s}{m_{iso}} - 1\right) \quad 2\text{-}1$$

$$G = \frac{m_d}{m_{iso}} \quad 2\text{-}2$$

where ρ_1 is the density of the swelling medium solvent; ρ_2 is the density of the composite material; m_s is the mass of the sample under swelling;

m_{iso} is the mass of the sample before swelling and extraction; $m_d$is the mass of the dry gel after the sample is dried after extraction.

3. Results and discussion

3.1. Characterization of F–SiO_2 and DHPM

As shown in Fig. 2(a), compared with the FTIR curve of SiO_2 there are two peaks appeared at 1621 cm^{-1} and 1401 cm^{-1} in the curve of F–SiO_2, which are corresponding to the stretching vibration absorption of the ester group and furan skeleton, respectively (see Fig. 1). As shown in Fig. 2(b), compare with the maleic anhydride (MAH), there is a broad stretching vibration of the hydroxyl group at 3500 cm^{-1} in the curve of DHPM. From 1340 cm^{-1} to 1436 cm^{-1} are corresponding to the deformation vibration of hydroxyl. In addition, at 1200 cm^{-1}-1000 cm^{-1} the peaks are attributed to the stretching vibration absorption peak of C–O. In addition, F–SiO_2 and the synthesis of F–SiO_2 and DHPM were further confirmed by ^{1}H-NMR spectroscopy as shown in Fig. 1S, Fig. 2S and Fig. 3S and 4S.

3.2. Thermal properties of MPU and FM

Fig. 3 (a) is the TGA diagram of the thermal decomposition curves of MPU and a series of FM. It can be found that almost all MPU and composites have two weightless platforms. The first weightless platform appeared at about 100 °C, which was mainly caused by the incomplete drying of the sample before the test. The moisture and impurities contained in it. The second weightless platform that appeared in the range of 250 °C–500 °C was the thermal decomposition of the materials themselves. On the whole, the filling of F–SiO_2 basically does not affect the thermal decomposition mechanism of the MPU matrix. In addition, the TGA curves of MPU and its composites exhibit two stages of thermal decomposition corresponding to the soft segment decomposition and hard segment decomposition, respectively. Specifically, the decomposition temperature of 10% weight loss ($T_{10\%}$) of MPU is higher than that of FM composites due to the inverse Diels-Alder reaction of the FM composites. And the $T_{10\%}$ is decreased with increasing in F–SiO_2 contents (see Fig. 4).

The glass transition temperature (Tg) is an important reference

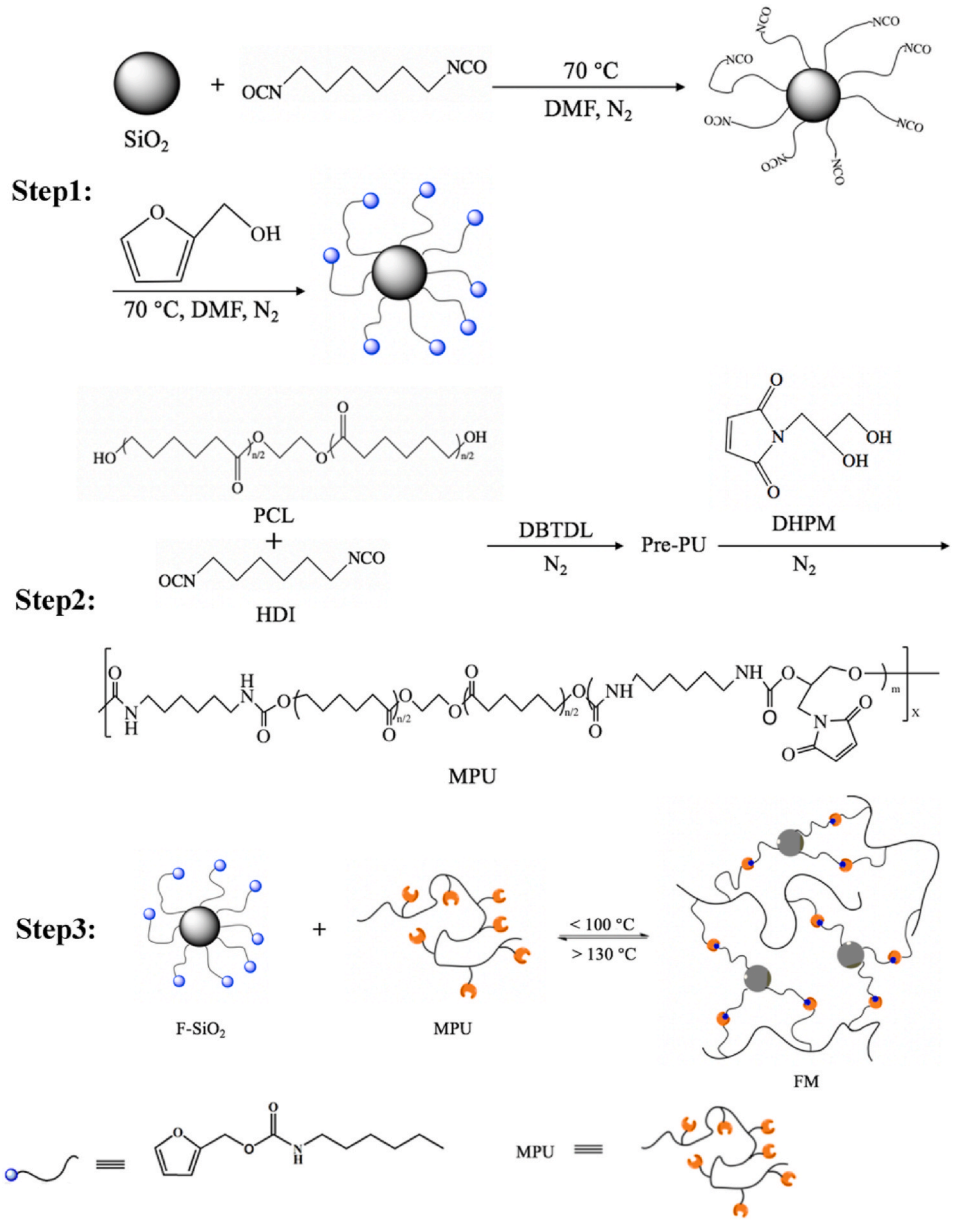


Fig. 1. The synthesis route of FM composites.

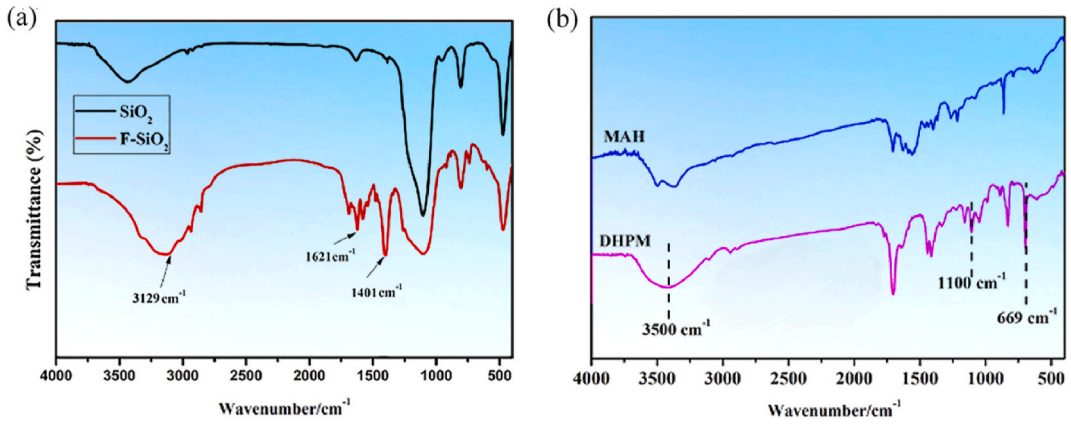


Fig. 2. (a) FTIR spectra of SiO_2 and F–SiO_2, (b) FTIR spectra of MAH and DHPM.

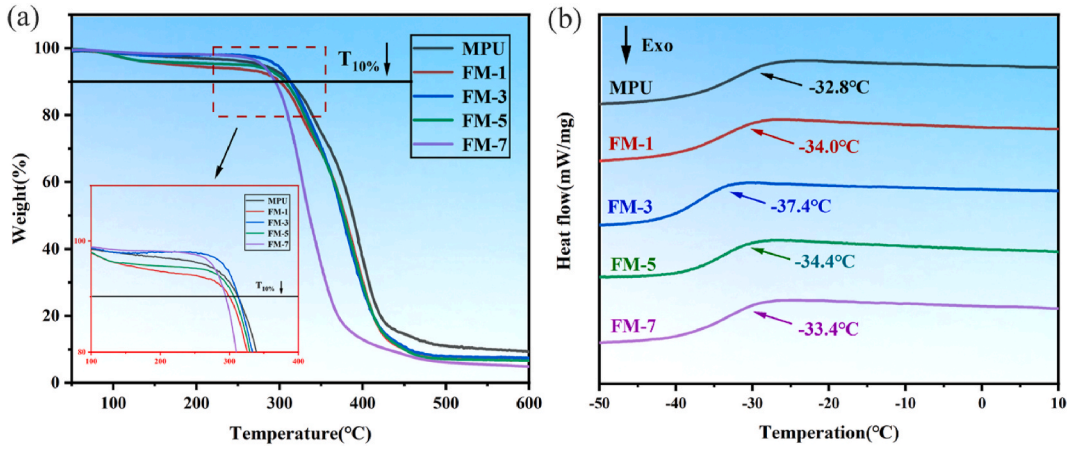


Fig. 3. The thermal properties of MPU and FM films (a) TGA, (b) DSC.

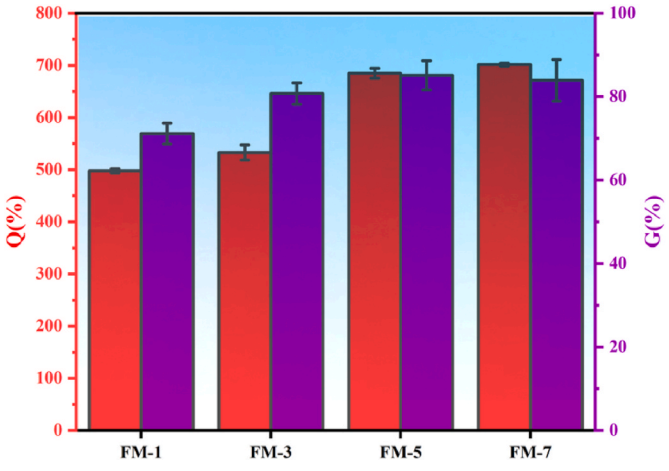


Fig. 4. Q and G of FM films.

factor that affects the practical application of material processing and manufacturing. The Tg of MPU, FM-1, FM-3, FM-5 and FM-7 were measured by differential scanning calorimetry, and the midpoint of the transition zone in the DSC curve was expressed as Tg. The result is shown in Fig. 3(b). The Tg of the films first decreased and then increased with the increase of F–SiO_2 addition. The Tg of MPU, FM-1, FM-3, FM-5 and FM-7 were −32.8 °C, −34.0 °C, −37.4 °C, −34.4 °C and −34.4 °C, respectively. The low glass transition temperature ensures that the internal molecular chain can still migrate at low temperatures, making it capable of being used at low temperatures. Therefore, from the analysis of TGA and DSC data, FM has excellent temperature stability.

3.3. Swelling properties of FM composites

The degree of crosslinking of the crosslinked network structure was studied by testing the degree of swelling (Q) and gel content (G). The Q and G values calculated according to formula 2-1 and 2-2 are listed in Table 2S. From the data in Table 2S it can be seen that as the proportion of F–SiO_2 filler increases, the Q value of FM materials gradually increased. Compared with FM-1, the swelling degree of FM-7 is increased by 203.67%. This might be because the double bond and amino structure on F–SiO_2 fully reacted with the corresponding structure on MPU, making the network cross-linking structure of the composite material more stable, and the molecular chain length is longer, so that more solvent enters the molecular chain.

In addition, it can be seen from the data in Table 2S that the addition of F–SiO_2 filler has little effect on the G value. The G value of each proportion of FM maintained between 70% and 90%. It shows that the network cross-linking structure of FM connects each molecular chain well, so only a small number of uncross-linked molecular chains will be washed away during extraction. The addition of F–SiO_2 filler makes the molecular chain of the FM films fully cross-linked, and the network cross-linked structure is more stable. This may be caused by the reaction between the double bond and amino structure introduced on the F–SiO_2 molecular chain and the structure on the MPU molecular chain.

3.4. Mechanical properties of MPU and FM

In order to study the effect of F–SiO_2 filler on the mechanical properties of MPU matrix, a series of MPU and FM dumbbell-shaped films were prepared. The specific data on mechanical properties are summarized in Table 3S. Fig. 5(b) shows the tensile stress and strain curves of pure MPU and FM at room temperature. It can be seen that the stress-strain curves of MPU and FM composites are typical elastic polymer curves and the entire stretching process can be divided into three regions. The first area is the elastic rise area at the initial stage. This area reflects the stiffness of the material. Obviously, with the increase of the F–SiO_2 filler content, the stiffness of the composite material has been significantly improved. The second area is the yield area of the elastic polymer. The increase in the F–SiO_2 filler content results in an increase in the yield stress. In the third area, the slope of the curve of each sample has a significant increase, and the strength of the samples are improved.

Fig. 5(a) characterized the tensile strength and elongation at break of the films. The tensile strength of the composite films increased with the increase of the F–SiO_2 filler content. It can be seen that the tensile strength of the unmodified MPU film was about 2.03 Mpa. When the content of F–SiO_2 is 7 wt%, the tensile strength of the composite material reached 15.54 MPa, which was 665.52% higher than that of pure MPU. This fully proved the reinforcing effect of F–SiO_2. However, the elongation at break showed a trend of first increasing and then decreasing. It is worth noting that the F–SiO_2 content of 5 wt% is the critical point for the growth of elongation at break. After that, the elongation at break of the film began to show a downward trend. As we all know, the elongation at break reflects the flexibility of the molecule. The greater the flexibility, the greater the elongation at break. Compared with pure MPU, the elongation at break of FM-5 was increased by 539.56%. Moreover, the elongation at break of pure MPU was only 306.05%, while the elongation at break of composites all exceed 500%. What's more, as shown in Fig. 6(a), taking the FM-3 film as an example, it can achieve bending, rolling, and stretching deformations, which further proves that it has good flexibility.

The above phenomenon is caused by the following aspects: First, the F–SiO_2 filler was introduced into two molecular structures of furan and maleic anhydride. Due to the D-A reaction between the conjugated diene and the monoene, the network of the composites was cross-linked, and the structure was more stable; however, an excessive amount of F–SiO_2 filler may cause agglomeration during the blending process, which will hinder the movement of the molecular chain during the stretching process, resulting in poor flexibility of the composites.

3.5. Output performance of F–SiO_2/MPU TENG (FM-TENG)

According to the triboelectric sequence, silicone rubber is a material with relatively high electronegativity, while polyurethane is a material with relatively low electronegativity. Therefore, this experiment used silicone rubber as the negative friction layer of FM-TENG, and FM films as the positive friction layer. The working mechanism of triboelectric nanogenerator is shown in Fig. 6(b). After the two materials were in contact, electrical energy was generated due to frictional electrification and electrostatic induction coupling. When silicone rubber and FM films frictionally contact, due to the triboelectric effect, the two materials will be charged with equal amounts of negative and positive charges. At this time, the electric field was in equilibrium. Silicone rubber has a strong ability to lose electrons, and FM-TENG has a strong ability to obtain electrons. Therefore, after friction contact, the surface of silicone rubber was positively charged, and FM-TENG belted negative electricity. When the external mechanical movement separated the two materials, a certain potential difference would occur between the two materials. The potential difference would increase as the separation distance increases until the two reach the maximum gap. If FM-TENG is connected to a load, electrons will be transferred from FM-TENG to the load due to the potential difference, forming a current in the external circuit. When the silicone rubber is in contact with FM-TENG again by an external force, the reduction of the gap will make the potential of the triboelectric nanogenerator lower than the load, and electrons will be transferred from the load to FM-TENG. There will also be a current in the opposite direction.

In order to test the output performance of the FM-TENG, we prepared a 2 × 2 cm^2 device for output testing and used a progressive motor to simulate the periodic contact and separation of the aluminum foil and the FM-TENG. The operating frequency was 1 Hz. As shown in Fig. 7, when the content of F–SiO_2 ranges from 1 wt% to 7 wt%, the short-circuit current (*Isc*), open circuit voltage (*Voc*) and transferred charge (*Qsc*) of the FM-TENG presented a trend of increasing first and then decreasing. Specifically, when the F–SiO_2 content is 3 wt%, the short-circuit current (*Isc*), open circuit voltage (*Voc*) and charge transfer amount (*Qsc*) reach their maximum values, which are approximately 1.81 μA, 69.61 V, and 23.942 nC, respectively. Compared with pure MPU, it has increased by 40.31%, 54.17% and 66.58%, respectively. The results show that the FM-TENG with 3 wt% F–SiO_2 can effectively convert mechanical energy into electrical energy, and exhibits excellent performance in terms of short-circuit current, open circuit voltage, and transferred charge.

As shown in Fig. 7(d), in order to characterize the output power of FM-TENG, we put a resistance of 10 kΩ–5 GΩ in series in the circuit. When the operating frequency was set to 1 Hz, the relationship between the short-circuit current density and the external load was obtained by testing the current at both ends of the resistor. At the same time, formula $P=I^2R$ was used to calculate the power density. It can be clearly seen that as the external load increases, the current density first decreases

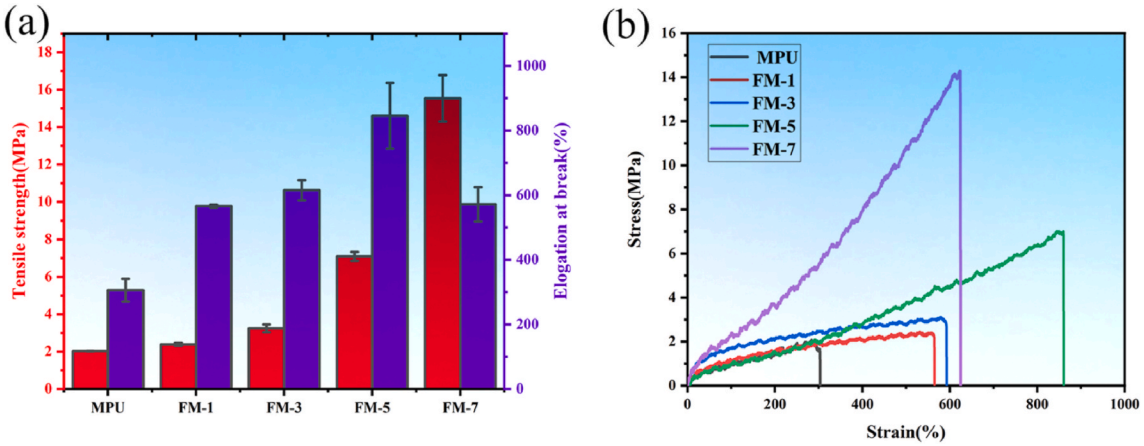


Fig. 5. The mechanical properties of MPU and FM films: (a) Tensile strength and elongation at break, (b) stress-strain curves.

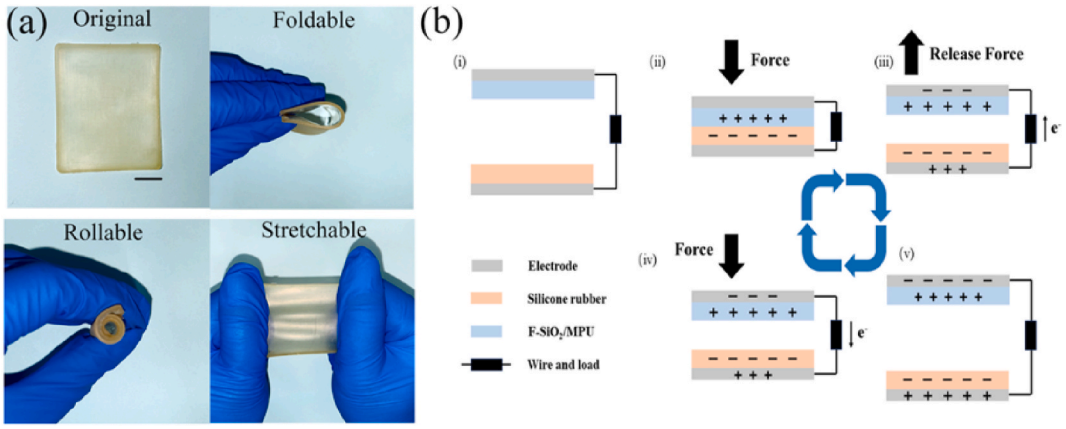


Fig. 6. (a) Photographs of the FM-3 film in its original, folding, rolling and stretching state (Scale bars, 1 cm), (b) the working mechanism of triboelectric nanogenerator.

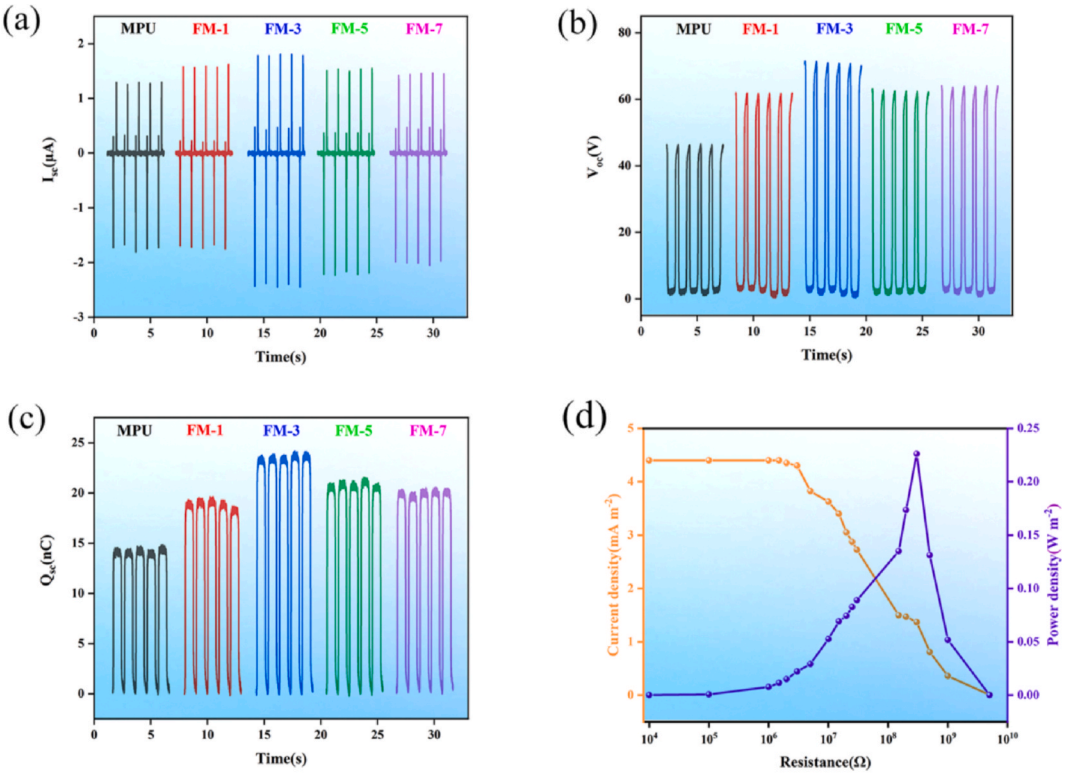


Fig. 7. The FM TENG at 1 Hz contact-separation frequency (a) short-circuit current (*Isc*), (b) open circuit voltage (*Voc*), (c) transferred charge (*Qsc*), (d) the relationship between peak power density and current density of FM TENG and the external load resistance.

slowly, and then decreases rapidly. While, as the load increases, the peak power density shows a trend of first increasing and then decreasing. When the external load is 300 MΩ, the peak power density reaches the maximum value of 0.226 W m^{-2}, which is 13.00% higher than MPU TENG (Fig. 5S). The results show that FM TENG can effectively convert mechanical energy into electrical energy.

3.6. Recyclable properties of FM composites

In order to prove the recyclability of the FM films, the FM films were dissolved in DMF solution (solid content is 10 wt%) and vacuum dried at 100 °C for 24 h. Take the FM-3 film as an example to illustrate the entire cycle in detail. At room temperature, the FM-3 film in DMF were insoluble and swells only slightly (Fig. 8 (a)), indicating the presence of a D-A cross-linked network. When heated to about 130 °C, the FM-3 film was completely dissolved, forming a transparent solution (Fig. 8 (b)), indicating that the D-A bond was broken due to the reverse D-A reaction. After vacuum drying at 100 °C for 24 h, the solution became a dry solid film again (Fig. 8 (c)). This process further proved that the current system is completely thermally reversible.

Fig. 8(d) shows the stress-strain curves of the original FM-3 film and the recycled FM-3 film. It can be seen that the two curves are almost identical. In addition, as shown in Fig. 8(e), it can be clearly seen that the reverse D-A reaction of FM films at around 100 °C. In summary, FM can be recycled.

The MPU, FM-1, FM-3, FM-5, and FM-7 films previously used for tensile testing were dissolved in DMF solution (Solid content is 10 wt%.) at 130 °C. Then, they were dried in a vacuum oven at 100 °C for 24 h to

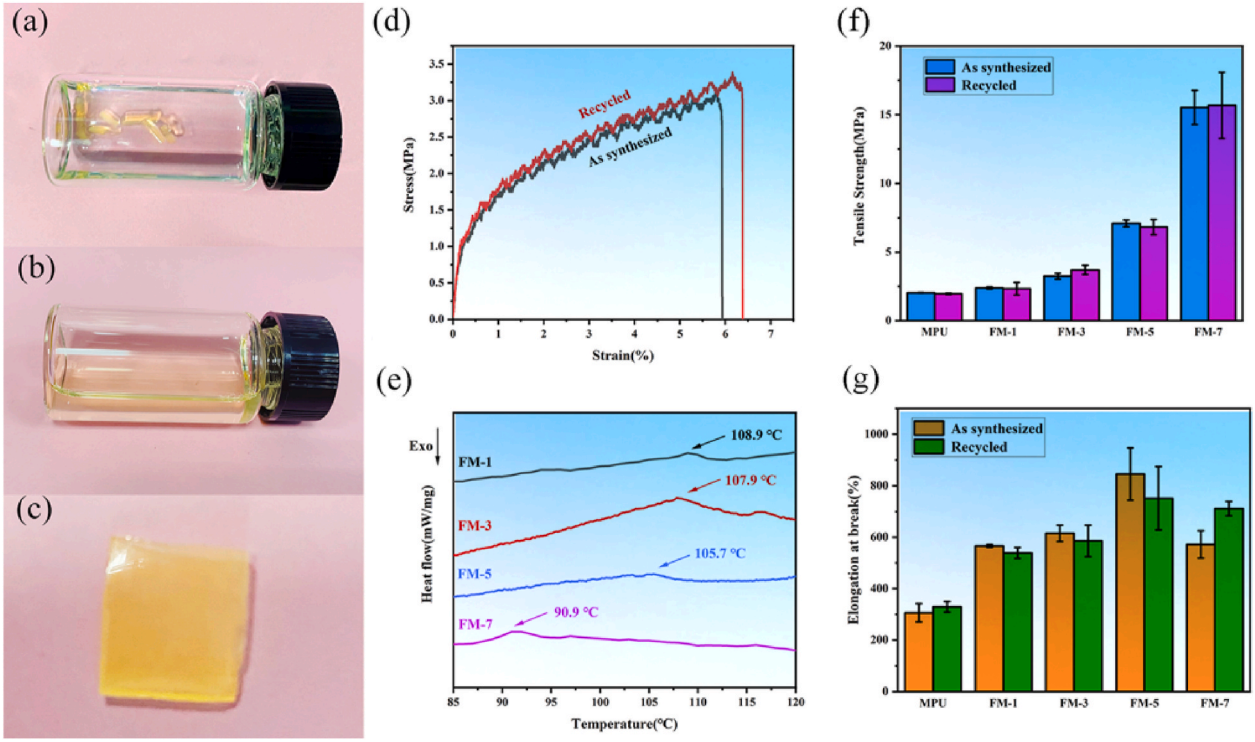


Fig. 8. (a) The FM-3 film swelled at room temperature, (b) the FM-3 film dissolved when heated for 5 min at 130 °C, (c) the FM-3 film redried in a vacuum oven for 24 h at 100 °C, (d) stress-strain curves of the original FM-3 film and recycled FM-3 film, (e) DSC curves of FM films, (f) the tensile strength of the original FM films and recycled FM films, (g) the elongation at break of the original FM films and recycled FM films.

form recycled films. The mechanical properties of the recycled films obtained by this method were further measured by a tensile test. The specific data are summarized in Table 4S. As shown in Fig. 8(f) and (g), the tensile strength and elongation at break of the recycled films were almost the same as the original films. The difference in tensile strength between recycled films and original films was 2.10%, 14.05%, 3.78% and 0.94%, respectively. The difference in elongation at break was 4.85%, 4.75%, 11.17% and 15.02%, respectively. In summary, the

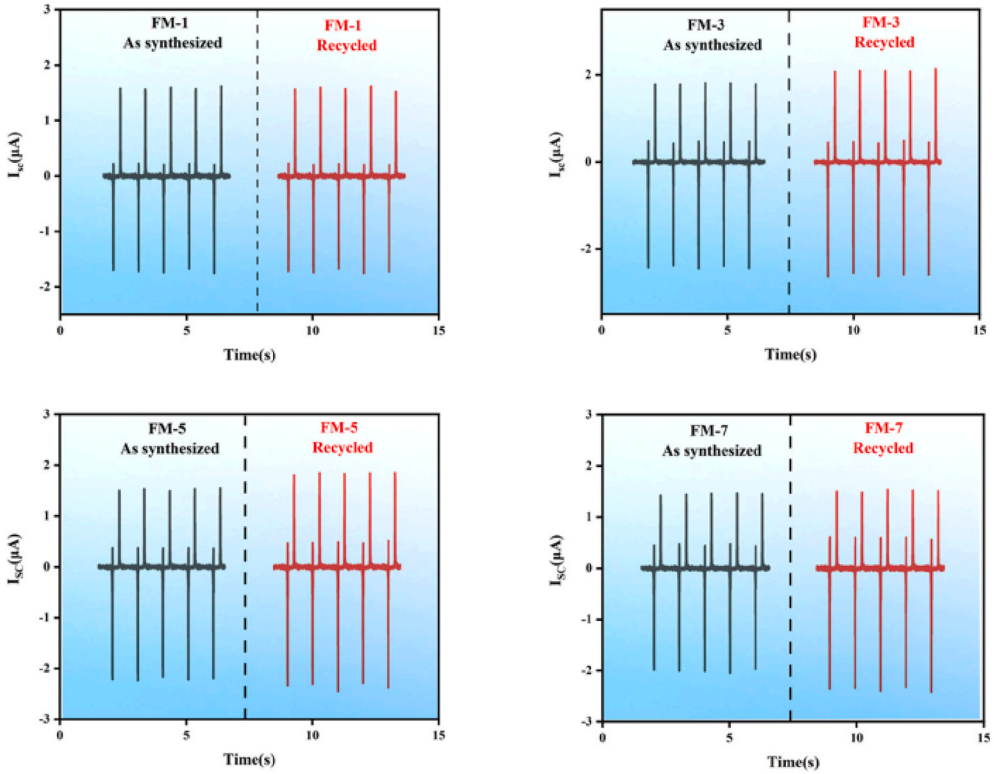


Fig. 9. The comparison of the short-circuit current (*Isc*) of FM films.

average recovery rates of the tensile strength and elongation at break of the recycled FM films are 94.78% and 91.05%.

Since the FM films are ultimately used to make TENG, the recyclability of its electrical properties is equally important. The electrical output performance of the recycled FM films obtained by vacuum drying again by the above method were also tested. Fig. 9, Fig. 10 and Fig. 11 respectively show the comparison of the short-circuit current (*Isc*), open circuit voltage (*Voc*) and transferred charge (*Qsc*) of the regenerated films with the corresponding data of the original films. It can be seen from Figs. 9, Figs. 10 and 11 that the short-circuit current (*Isc*), open circuit voltage (*Voc*) and transferred charge (*Qsc*) data of the regenerated films were basically consistent with the corresponding data of the original film. Among them, the short-circuit current (*Isc*) difference is 0.03%, 14.91%, 20.26% and 5.51%. The open circuit voltage gap (*Voc*) is 0.22%, 8.93%, 0.60% and 3.58%. The difference in charge transfer amount (*Qsc*) is 12.91%, 7.84%, 1.02% and 16.23%. To sum up, the average recovery rates of short-circuit current (*Isc*), open-circuit current (*Voc*) and transferred charge (*Qsc*) between the original films and recycled films are 89.82%, 96.67%, and 90.50%. This demonstrated that the recyclability of the electrical properties of the FM films was also excellent.

3.7. Application of FM-TENG

As shown in the Supporting movie, FM-TENG can be used as a self-charging power source to power LED bulbs. Connect the FM-TENG (the content of F–SiO_2 of the FM film is 3 wt%) with 20 small white LED bulbs in series (Fig. 12(a)), and then tap the FM-TENG with palms. 20 small LED bulbs can be lit. Another important application of FM-TENG is as a self-powered sensor to monitor the movement of the human body in real time. As shown in Fig. 12(b), a device with a hydrogel inside and an FM-3 film wrapped outside was attached to the joint of the finger. Different voltage signals are generated by the different amplitudes of finger bending. When the finger is straightened, the voltage was 0. When the finger is slightly bent, the voltage signal increased to 2.30 V. When the finger is dramatically bent, the voltage signal increased to 5.31 V. Therefore, we can judge the angle of finger bending based on the magnitude of the voltage signal generated.

4. Conclusions

In this study, the two molecular structures of maleic anhydride and furan were respectively introduced into the polyurethane matrix and the surface of SiO_2. Utilizing the Diels-Alder reaction between conjugated diene and monoene, the synthesized composites possessed both thermosetting and thermoplastic polymer properties. In addition, the interface compatibility between MPU and F–SiO_2 was also improved, thus obtaining a nano-SiO_2 enhanced reversible conversion material. Moreover, FM composites has excellent mechanical properties, electrical output properties, and thermal recyclability. When F–SiO_2 content is 5 wt% (FM-5), the tensile strength and elongation at break are 7.09 MPa and 845.61%, respectively. Compared with pure MPU, it has increased by 249.26% and 176.30%, respectively. At the same time, the electrical output performance of FM-3 is excellent. The short-circuit current, open circuit voltage, transferred charge and the peak power density of FM-TENG with 3 wt% F–SiO_2 are 1.53 μA, 61.42V and 21.22 nC and 0.226 W m^{-2}, respectively. Compared with pure MPU, it has increased by 18.60%, 36.04%, 49.24% and 13.00%, respectively. In terms of application, 20 LED bulbs can be lighted by continuously tapping the FM-TENG device with palms. At the same time, FM-TENG can also be used as a self-powered sensor to monitor finger movement.

Furthermore, after the original films were re-dissolved and vacuum dried, the mechanical properties and electrical output performance were tested. The average recovery rates of the tensile strength and elongation at break of the recycled FM films are 94.78% and 91.05%. Additionally, the average recovery rates of short-circuit current (*Isc*), open-circuit current (*Voc*) and transferred charge (*Qsc*) between the original films and recycled films are 89.82%, 96.67%, and 90.50%, respectively.

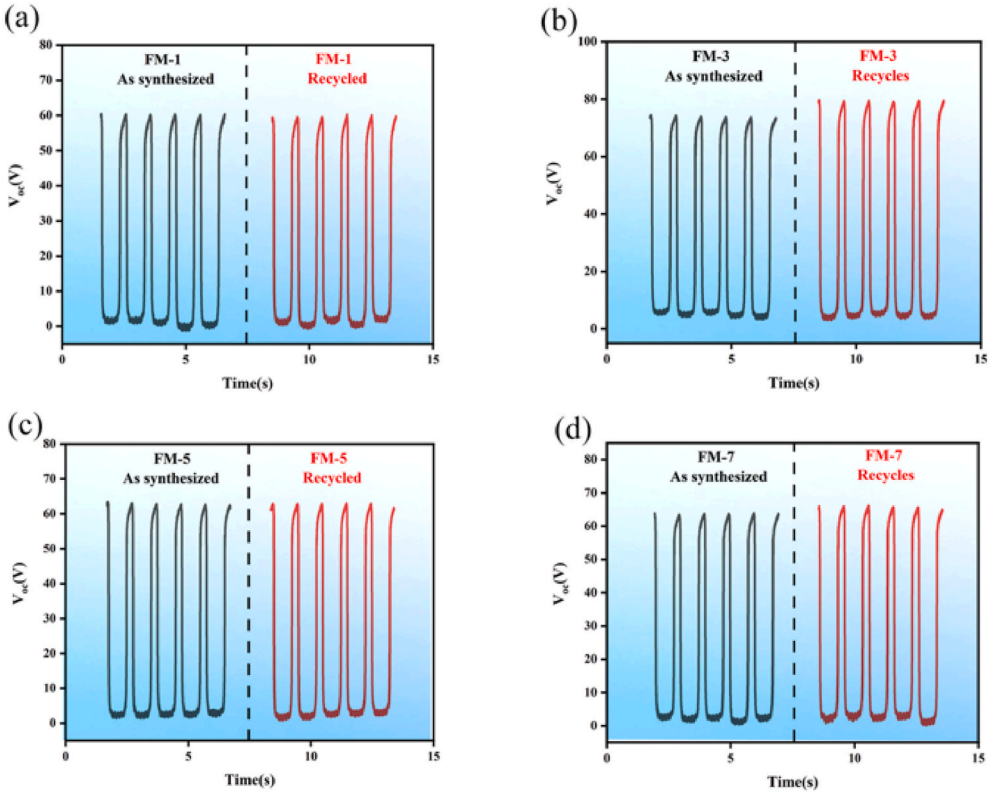


Fig. 10. The comparison of the open circuit voltage (*Voc*) of FM films.

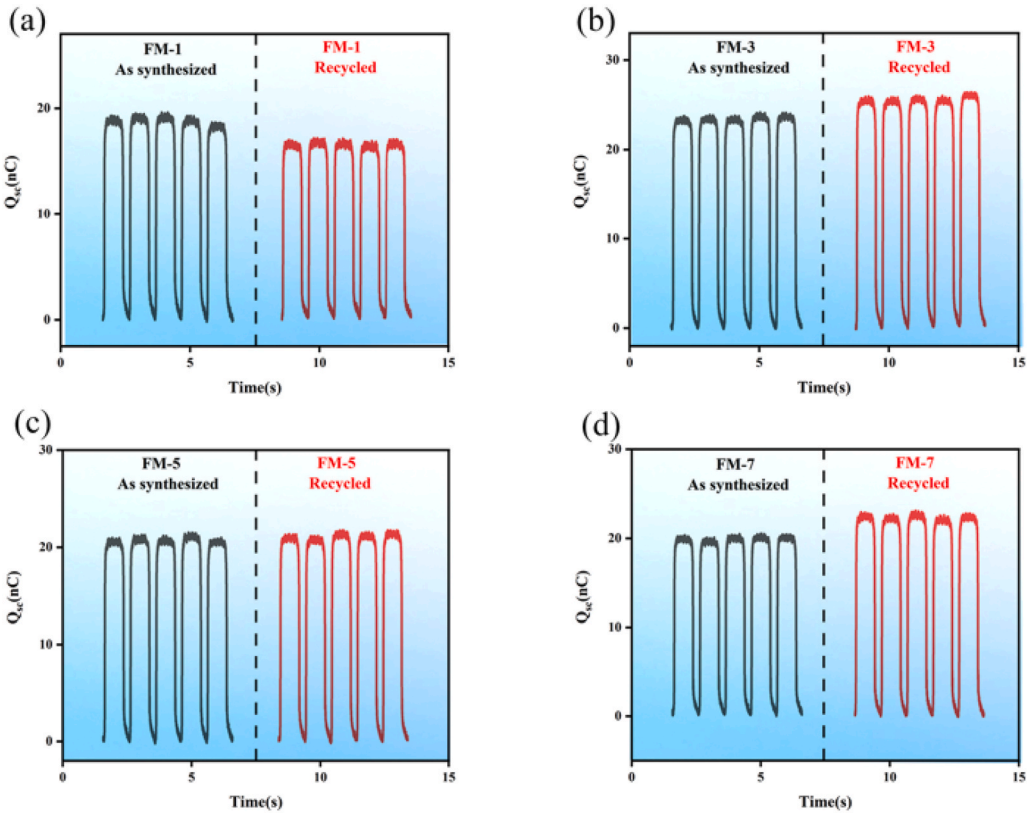


Fig. 11. The comparison of the transferred charge (*Qsc*) of FM films.

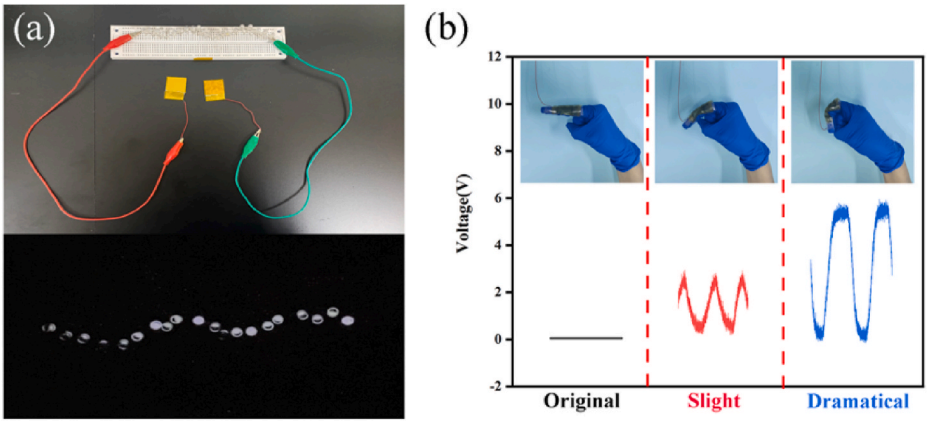


Fig. 12. (a) Directly light up 20 LED lights by tapping the FM-TENG with your palm, (b) FM-TENG is used to monitor the finger bending amplitude.

Therefore, FM composites has excellent recyclability. In sum, flexible and recyclable FM composites base TENG can be used as a self-driving power supply device for a variety of portable electronic devices.

Credit author statement

Xu Xu: Formal analysis, Writing- Original draft preparation, Resources. Yuting Ouyang: Investigation, Writing-Review & Editing. Hong Ruan: Data curation. Shaorong Lu: Writing-Reviewing and Editing. Yuqi Li: Conceptualization, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support of the National Science Foundation of China (51605109 and 51763009), the Guangxi Natural Science Foundation (2018GXNSFBA281052 and 2018GXNSFAA281296), Young teachers, basic ability to improve project of Guangxi Province (2018KY0243), State Key Laboratory of Bio-Fibers and Eco-Textiles (KF2020203), Scientific Research and Technology Development Plan of Guilin, China (20190219-4) and the Project of Thousand Outstanding Young Teachers' Training in Higher Education Institutions of Guangxi.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compscitech.2021.109068.

References

[1] X. Li, C. Jiang, F. Zhao, et al., Fully stretchable triboelectric nanogenerator for energy harvesting and self-powered sensing, Nanomater. Energy 61 (2019) 78–85, https://doi.org/10.1016/j.nanoen.2019.04.025.

[2] F.R. Fan, W. Tang, Z.L. Wang, Flexible nanogenerators for energy harvesting and self-powered electronics, Adv. Mater. 28 (2016) 4283–4305, https://doi.org/10.1002/adma.201504299.

[3] W. Gao, S. Emaminejad, H. Y, Y. Nyein, et al., Fully integrated wearable sensor arrays for multiplexed in. situ perspiration analysis, Nature 529 (2016) 509–514, https://doi.org/10.1038/nature16521.

[4] Z.L. Wang, Commercial application of nano generators as sustainable power sources and active sensors, Sci. China Chem. 43 (2013) 759–762.

[5] Z. Wen, M.-H. Yeh, H. Guo, et al., Self-powered textile for wearable electronics by hybridizing fiber-shaped nanogenerators, solar cells, and supercapacitors, Science Advances 2 (2016), e1600097, https://doi.org/10.1126/sciadv.1600097.

[6] Z.L. Wang, Triboelectric nanogenerators as new energy Technology for self-powered systems and as active mechanical and chemical sensors, ACS Nano 7 (2013) 9533–9557, https://doi.org/10.1021/nn404614z.

[7] F. Fan, Z.Q. Tian, Z. Wang, Flexible triboelectric generator, Nanomater. Energy 1 (2012) 328–334, https://doi.org/10.1016/j.nanoen.2012.01.004.

[8] Y. Zi, S. Niu, J. Wang, et al., Standards and figure-of-merits for quantifying the performance of triboelectric nanogenerators, Nat. Commun. 6 (2015) 8376, https://doi.org/10.1038/ncomms9376.

[9] R. Zhang, S. Wang, M.-H. Yeh, et al., A streaming potential/current-based microfluidic direct current generator for self-powered nanosystems, Adv. Mater. 27 (2015) 6482–6487, https://doi.org/10.1002/adma.201502477.

[10] Q. Liang, X. Yan, X. Liao, et al., Multi-unit hydroelectric generator based on contact electrification and its service behavior, Nanomater. Energy 16 (2015) 329–338, https://doi.org/10.1016/j.nanoen.2015.07.004.

[11] P.-C. Hsu, A.Y. Song, P.B. Catrysse, et al., Radiative human body cooling by nanoporous polyethylene textile, Science 353 (2016) 1019–1023, https://doi.org/10.1126/science.aaf5471.

[12] K.N. Kim, J. Chun, J.W. Kim, et al., Highly stretchable 2D fabrics for wearable. Triboelectric nanogenerator under harsh environments, ACS Nano 9 (2015) 6394–6400, https://doi.org/10.1021/acsnano.5b02010.

[13] Z.L. Wang, Nanogenerators, self-powered systems, blue energy, piezotronics and. piezo-phototronics – a recall on the original thoughts for coining these fields, Nanomater. Energy 54 (2018) 477–483, https://doi.org/10.1016/j.nanoen.2018.09.068.

[14] Z.L. Wang, Principles and potential applications of piezoelectric nanometer generators, Physical 11 (2006) 897–903.

[15] P. Lv, L. Shi, C. Fan, et al., Hydrophobic ionic liquid gel-based triboelectric nanogenerator: next generation of ultrastable, flexible, and transparent power sources for sustainable electronics, ACS Appl. Mater. Interfaces 12 (2020) 15012–15022, https://doi.org/10.1021/acsami.9b19767.

[16] S. Wang, L. Lin, Z.L. Wang, Triboelectric nanogenerators as self-powered active sensors, Nanomater. Energy 11 (2015) 436–462, https://doi.org/10.1016/j.nanoen.2014.10.034.

[17] W. Zheng, Y.Q. Yang, N. Sun, et al., A wrinkled PEDOT:PSS film based stretchable and transparent triboelectric nanogenerator for wearable energy harvesters and active motion sensors, Adv. Funct. Mater. 28 (2018), https://doi.org/10.1002/adfm.201803684.

[18] Z.L. Wang, On Maxwell's displacement current for energy and sensors: the origin of, Nanogenerators, *Materials Today* 20 (2017) 74–82, https://doi.org/10.1016/j.mattod.2016.12.001.

[19] F. Fringuelli, A. Taticchi, The Diels–Alder Reaction, Selected Practical Methods, Chichester, 2001, 343-340.

[20] H. Nandivada, X. Jiang, J. Lahann, Click chemistry: versatility and control in the hands of materials scientists, Adv. Mater. 19 (2007) 2197–2208, https://doi.org/10.1002/adma.200602739.

[21] J.H. Li, G.P. Zhang, R. Sun, et al., Self-healing polymer material based on Diels-Alder Reaction, Polym. Bull. (2014) 8–15.

[22] M. Watanabe, N. Yoshie, Synthesis and properties of readily recyclable polymers from bisfuranic terminated poly(ethylene adipate) and multi-maleimide linkers, Polymer 47 (2006) 4946–4952, https://doi.org/10.1016/j.polymer.2006.05.036.

[23] C. Goussé, A. Gandini, Synthesis of 2-furfurylmaleimide and preliminary study of its Diels-Alder polycondensation, Polym. Bull. 40 (1998) 389–394, https://doi.org/10.1007/s002890050267.

[24] S. Yu, R.C. Zhang, Q. Wu, et al., Bio-inspired high-performance and recyclable cross-linked polymers, Adv. Mater. 25 (2013) 4912–4917, https://doi.org/10.1002/adma.201301513.